

Experience

Unity 3D Engine • 10+ years • PC/Mobile development
Unreal Engine • 10+ years • Simulation development
Computer Vision • 10+ years • AI/ML, rendering development
GPGPU/CUDA • 10+ years • High performance computing

C/C++ • 20+ years • High perf systems, blockchains
Rust • 10+ years • Blockchain, AI, simulation
Python • 20+ years • Tools and pipeline engineering
F#/C# • 10+ years • Apps, games and tools

Employment

- Shield AI** <https://shield.ai/> **Principal AI and Simulation Engineer** **2018-**
- AI and simulation infrastructure, harness, training, synthetics and building world models where humans, robots and AI agents learn, play and evolve together.
 - Developed the simulator shown in the Hivemind fixed-wing autonomy demo.
 - Developing synthetic datasets for training ML models and virtual environments for training robots.
 - Developing scalable simulation systems for robot swarms using Unreal Engine, PX4, ROS (Robot Operating System)
 - Engineering autonomous AI agents for tactical mission planning, scenario generation, and CONOPS environments.
- Vertex Studio** <https://vrz.group> **Founder & CTO** **2016-2026**
- Building scalable simulation infrastructure solutions and immersive experiences centered on human-robot-AI interaction.
 - Designing and implementing multi-agent frameworks and behavior systems.
 - Developing deep learning pipelines using PyTorch, real time pipelines using GStreamer and simulations using Unreal Engine, Unity 3D and custom engines.
 - Developing core blockchains for scaling protocol-owned economies for games, simulations and XR experiences. Smart contracts, DeFi, GameFi, market makers.
 - Maintaining custom Unreal Engine + NVidia PhysX for robotics applications.
- Simbotic AI** <https://simbotic.ai/> **Lead Simulation and AI Engineer** **2018-2022**
- Leading development of open source physical AI platform for sim-to-real transfer and digital twin applications.
 - Engineering real-time synthetic data pipelines for training embodied AI models.
 - Integrating physics-based dynamics and computer vision into game engines.
 - Improving sample efficiency of reinforcement learning models with rich virtual environments with dense rewards.
 - Maintaining several docker containers for CUDA, ROS, GStreamer and TensorFlow.
 - Custom 3D engine development for vectorizing meta-layers for XR experiences.
- Civil Maps** <https://civilmaps.com/> **Simulation Consultant** **2017-2019**
- Velodyne** <https://velodynelidar.com/>
- Galois** <https://galois.com/>
- Sensor characterization, customization and visualization tools.
 - Integration of Unreal Engine into development operations.
 - Massive autonomous swarming simulation architecture for DARPA, DoD and similar agencies.
- Red Pill VR** <http://redpillvr.com/> **Systems Engineer & Technical Art** **2016-2017**
- Integrated real time deep learning inference for VR Music MMO game using TensorFlow, GStreamer and UE4.
 - Physics-based VR gameplay, UX interaction and full-body VR networking.
- Rawbots** <https://rawbots.github.io/> **Indie Game Developer** **2015-2016**
- Designed and develop Rawbots, a robot crafting sandbox game. Robot simulations developed with Bullet Physics.
- LucasArts & ILM** <https://www.lucasfilm.com/> **AAA Game Developer** **2005-2015**
- MunkyFun**
- Beyond Games**
- Console game developer for Xbox, PlayStation, Nintendo, PC, iOS and Android.
 - Gameplay, physics and network developer for multiple titles, most notable Star Wars: The Force Unleashed.
 - Simulation tools for real-time special effects and pre-vis for several movies.

Skills

C • C++ • Rust	ROS • PX4	Console • PC • Mobile	Git • Github • Gitlab
F# • C#	TensorFlow GPU	OpenGL • CUDA • GLSL	Win • Linux • Android • iOS
Python	GStreamer	THREE.js • A-Frame	Docker • AWS • GCP
Type/JavaScript	UnrealEngine • Unity3D	LLMs • Vision	TCP UDP WebRTC WebSockets
Erlang	Bullet Physics • PhysX	Nvidia Xavier/Jetson	Make • CMake • Bazel